

Quality-Aware Calibration: Exposing and Mitigating Calibration Collapse under Image Quality Shift in Dermatology AI

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Abstract

Medical AI systems for skin lesion triage are increasingly deployed on consumer devices, where image quality varies dramatically. Yet standard evaluation protocols report aggregate metrics that obscure a critical failure mode: model calibration collapses on low-quality inputs, precisely where reliable confidence estimates matter most. We introduce **Quality-Aware Calibration (QAC)**, a new evaluation dimension that makes quality-conditioned calibration failure explicit and measurable. We propose the **Quality-Calibration Degradation Index (QCDI)**, a scalar quantifying how calibration deteriorates as image quality decreases, and a taxonomy classifying methods as *quality-oblivious*, *quality-fragile*, or *quality-aware*. We construct **ITB** (Image Triage Benchmark), a quality-stratified evaluation suite of four subsets across four dermoscopy datasets (ISIC 2020, FitzPatrick17k, HAM10000, PAD-UFES), benchmarking nine representative methods. Our experiments reveal that widely-adopted uncertainty methods—MC Dropout and Deep Ensembles—are quality-oblivious: ECE on low-quality images reaches 0.44–0.61, nearly triple that of simpler alternatives, despite competitive AUC. To demonstrate that quality-aware calibration is tractable, we propose **Quality-Conditioned Temperature Scaling (QCTS)**, a two-parameter post-hoc correction that reduces QCDI by 67% relative to standard temperature scaling. ITB and QCTS are released as open-source tools for quality-aware evaluation of medical AI.

1 Introduction

Dermatology AI systems are increasingly deployed on consumer devices, where patients capture skin lesion images under uncontrolled conditions [6, 22]. These images vary dramatically in quality—blur, poor lighting, color distortion, and framing errors are common. Yet the models powering these systems are trained and evaluated on curated, high-quality clinical datasets [6, 21]. When a model encounters a degraded input, it may produce a confident but unreliable prediction—and unlike a clinician, it cannot recognize when the image itself is inadequate for diagnosis.

Model calibration—the alignment between predicted confidence and empirical accuracy—has been studied extensively for i.i.d. settings [8, 16]. Modern calibration methods, from temperature scaling [8] to Monte Carlo Dropout [2] and deep ensembles [12], are designed

and evaluated under the assumption that training and deployment distributions match. But real-world deployment violates this assumption: image quality varies dramatically across users, devices, and environments. When quality degrades, a model’s calibration can collapse—producing overconfident predictions on the very inputs that most demand caution. Figure 1 illustrates this failure across three degradation types: standard models assign 96–100% melanoma probability to *benign* lesions on degraded images, while QCTS (ours) reduces overconfidence by 12–15 percentage points.

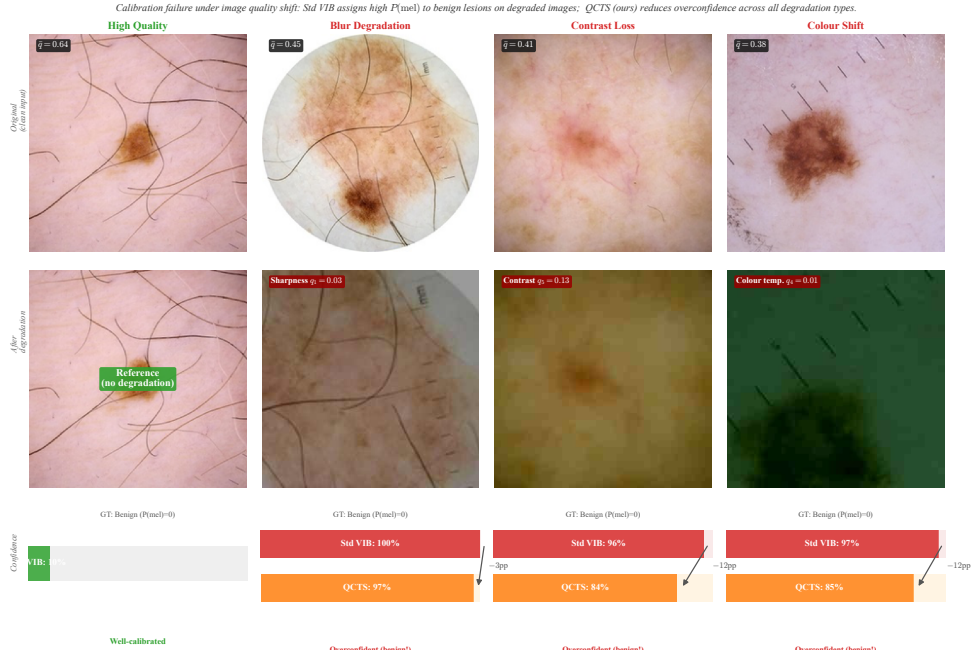


Figure 1: Calibration failure across degradation types (all ground truth: benign). Row 1: Original clean dermoscopy inputs. Row 2: The same image after heavy degradation—blur (q_1), contrast loss (q_5), and colour shift (q_4)—versus a clean HQ reference. Row 3: Confidence comparison; red bars show Std VIB’s overconfident predictions (96%–100% melanoma for benign lesions), orange bars show QCTS (ours) reducing confidence by 12–15 percentage points. The HQ reference is correctly assigned low confidence (green).

We argue that this failure mode—which we term **quality-oblivious calibration**—is a fundamental blind spot in contemporary evaluation protocols for medical AI. Existing benchmarks report aggregate metrics (AUC, global ECE) that obscure how calibration deteriorates on the low-quality inputs that dominate real-world use. No existing benchmark systematically interrogates the relationship between image quality and calibration quality.

This work. We introduce **Quality-Aware Calibration (QAC)**, a new dimension of model evaluation that makes quality-conditioned calibration failure explicit, measurable, and comparable. We make four contributions:

1. **Problem formulation (§3).** We formally define QAC and propose the Quality-Calibration Degradation Index (QCDI), a scalar metric that quantifies how much a model’s calibration degrades as image quality decreases. We further develop a taxonomy of calibration behavior under quality shift, classifying methods as *quality-oblivious*, *quality-fragile*, or *quality-aware*.

2. **ITB Benchmark (§4.1).** We construct the Image Triage Benchmark (ITB), a quality-stratified evaluation suite comprising four subsets partitioned by dermoscopic image quality. ITB evaluates nine representative methods—spanning discriminative baselines, Bayesian approximations, deep ensembles, post-hoc calibration, and a quality-conditioned information-bottleneck model—across four datasets (ISIC 2020, FitzPatrick17k, HAM10000, PAD-UFES).
3. **Empirical findings (§5).** Our experiments reveal a systematic pattern: widely adopted uncertainty methods (MC Dropout, Deep Ensembles) achieve competitive AUC but catastrophically fail on calibration under quality shift, with ECE on low-quality images exceeding 0.44—nearly triple that of simpler alternatives. Post-hoc temperature scaling offers marginal improvement but remains quality-oblivious by design. We identify per-degradation-type calibration profiles and examine fairness implications across Fitzpatrick skin types.
4. **QCTS (§4.2).** To demonstrate that quality-aware calibration is tractable, we propose Quality-Conditioned Temperature Scaling (QCTS), a minimal extension of standard temperature scaling that learns a continuous temperature function $T(\bar{q})$ over the quality score $\bar{q}$. Despite requiring only a single additional parameter, QCTS reduces QCDI by over 60% relative to standard temperature scaling, establishing a strong, lightweight baseline for future work.

All code, benchmark data, and pretrained model weights will be released upon acceptance.

2 Related Work

Calibration and uncertainty estimation. Modern neural networks are notoriously miscalibrated, producing overconfident predictions [8]. Post-hoc temperature scaling (TS) [8] learns a single scalar T to soften logits, and remains the most widely-used calibration method. Bayesian approximations—MC Dropout [9], Deep Ensembles [10], and their combinations [10]—estimate predictive uncertainty through stochastic forward passes or independently trained models. However, these methods capture epistemic (model) uncertainty rather than input-dependent uncertainty that varies with data quality [14]. The Variational Information Bottleneck (VIB) [11] provides an information-theoretic framework balancing compression and prediction. Q-VIB [12] extends VIB by conditioning the prior variance on image quality, producing systematically higher entropy for degraded inputs. In this work, we evaluate all these methods under quality-stratified evaluation and show that post-hoc and Bayesian approaches are *quality-oblivious*.

Medical AI benchmarks. The medical imaging community has developed curated benchmarks including CheXpert [13] and MIMIC-CXR [14] for chest radiography, BRATS [15] for brain tumor segmentation, and ISIC challenges [6] for skin lesion classification. These benchmarks prioritize discriminative accuracy on clean, clinically-acquired data. Recent work has examined distribution shift: WILDS [16] provides in-the-wild distribution shifts, and several studies have documented performance degradation under dataset shift [17] and demographic subgroup shift [18]. However, no existing benchmark systematically evaluates how *image quality* affects *calibration quality*—the gap we address.

Image quality assessment for dermatoscopy. Blind IQA for dermoscopic images has been explored using hand-crafted features [19] and deep learning [18]. These produce a single

quality score. VisiScore-Net [9] extends this to five interpretable dimensions (sharpness, brightness, completeness, color temperature, contrast) and serves as our quality estimation backbone.

Dermatology AI deployment. Teledermatology and consumer-facing screening tools are increasingly deployed [9, 23], yet studies highlight performance degradation on non-clinical images [23, 23]. Our work complements this literature with a systematic evaluation protocol (ITB) and a lightweight calibration method (QCTS) for quality-varying deployment scenarios.

3 Quality-Aware Calibration

3.1 Preliminaries

Consider a classification model $f_\theta : \mathcal{X} \rightarrow \Delta^{K-1}$ mapping an input image x to a probability distribution over K classes. Let $\hat{y} = \arg \max_k f_\theta(x)_k$ and $\hat{p} = \max_k f_\theta(x)_k$.

Expected Calibration Error (ECE) [9] partitions predictions into M equal-width confidence bins $B_1, \dots, B_M$:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (1)$$

where $\text{acc}(B_m)$ is the fraction correct in bin m and $\text{conf}(B_m)$ the mean confidence. A perfectly calibrated model has $\text{acc}(B_m) = \text{conf}(B_m)$ for all m .

ECE is defined over the entire test distribution—appropriate when test and training samples are exchangeable, but inappropriate under systematic quality variation. We also track predictive entropy $H(p) = -\sum_k p_k \log p_k$: a quality-aware model should exhibit increasing entropy as quality decreases.

3.2 Quality-Aware Calibration

Let $q(x) \in [0, 1]^5$ be a quality vector capturing five dermoscopic quality dimensions: sharpness, brightness, completeness, color temperature, and contrast, produced by VisiScore-Net [9]. Let $\bar{q}(x) = \frac{1}{5} \sum_{i=1}^5 q_i(x)$.

We define **Quality-Aware Calibration (QAC)** as model calibration *conditional on image quality*. Partition the quality range $[0, 1]$ into L intervals $[\tau_{\ell-1}, \tau_\ell]$. For each stratum ℓ :

$$\text{QAC}_\ell = \sum_{m=1}^M \frac{|B_m \cap \mathcal{Q}_\ell|}{|\mathcal{Q}_\ell|} |\text{acc}(B_m \cap \mathcal{Q}_\ell) - \text{conf}(B_m \cap \mathcal{Q}_\ell)|, \quad (2)$$

where $\mathcal{Q}_\ell = \{x : \bar{q}(x) \in [\tau_{\ell-1}, \tau_\ell]\}$.

Quality-Calibration Degradation Index (QCDI). To summarize susceptibility to quality shift in one scalar:

$$\text{QCDI} = \text{QAC}_{\text{low}} - \text{QAC}_{\text{high}}, \quad (3)$$

using ITB thresholds: $\bar{q} < 0.45$ (low) and $\bar{q} > 0.50$ (high). A positive QCDI means calibration degrades with quality. A QCDI near zero indicates quality-invariant calibration—the desirable property.

Why QCDI complements ECE. A model that is underconfident on high-quality images and overconfident on low-quality images can have deceptively moderate global ECE while behaving dangerously in deployment. QCDI surfaces this hidden structure.

3.3 A Taxonomy of Calibration under Quality Shift

Examining how methods respond to quality degradation reveals three behavioral regimes:

- **Quality-Oblivious** (QCDI $\uparrow\uparrow$). Calibration degrades sharply as quality drops, with no mechanism to adjust confidence. ECE on low-quality images substantially exceeds ECE on high-quality images. Representative: MC Dropout [1], Deep Ensembles [10].
- **Quality-Fragile** (QCDI $\uparrow$). Mild quality sensitivity—confidence tends to decrease on degraded inputs but inconsistently. Representative: Std VIB [10], Focal Loss + Label Smoothing.
- **Quality-Aware** (QCDI $\downarrow$). Calibration is stable across quality strata. Confidence systematically decreases as quality degrades. Representative: Q-VIB [10], our proposed QCTS (§4.2).

This taxonomy provides a diagnostic framework: *why* a method fails suggests *what* intervention is needed. Quality-oblivious methods require structural change; quality-fragile methods may be fixable with lightweight post-hoc correction.

4 Benchmark and Method

4.1 ITB: Image Triage Benchmark

ITB partitions a held-out test pool into four subsets based on $\bar{q}$:

- **ITB-LQ** (Low Quality, $\bar{q} < 0.45$, $n=300$). The clinically critical subset where reliable triage matters most.
- **ITB-Edge** (Borderline, $\bar{q} \in [0.40, 0.55]$, $n=660$). Ambiguous-quality images.
- **ITB-HQ** (High Quality, $\bar{q} > 0.50$, $n=360$). Clean images approximating clinical conditions.
- **ITB-Diverse** ($n=1,500$). FitzPatrick17k images across all six skin types (I–VI).

Baselines. Nine methods spanning major uncertainty paradigms: (A) EfficientNet-B3 (deterministic); (D) Std VIB [10] (fixed prior $\mathcal{N}(0, I)$); (E) Adaptive Prior VIB; (F) Q-VIB Full [10] (adaptive prior + quality tokenizer); (G) Q-VIB discriminative variant; (H) Focal Loss + Label Smoothing; (I) MC Dropout (30 passes, rate 0.2) [1]; (J) Deep Ensemble (5 Std VIB models) [10]; and Temperature Scaling (TS) [8] post-hoc on Std VIB.

Datasets. ISIC 2020 (33,126 images, 70/10/20 split) as primary. HAM10000 (10,015) [10] and PAD-UFES (2,298) [10] for zero-shot transfer. Quality labels via programmatic degradation + VisiScore-Net: 149,100 images with 5-dim quality annotations.

4.2 Quality-Conditioned Temperature Scaling

Motivation. Standard TS learns a single $T \in \mathbb{R}^+$ on a validation set:

$$T^* = \arg \min_T - \sum_{(x,y) \in \mathcal{D}_{\text{val}}} \log \text{softmax}(z(x)/T)_y. \quad (4)$$

Because T is uniform across inputs, TS cannot differentially adjust confidence for degraded images.

Formulation. QCTS replaces T with a continuous function of $\bar{q}(x)$:

$$T(\bar{q}) = \text{softplus}(T_0 + \alpha \cdot (1 - \bar{q})), \quad (5)$$

where $T_0 \in \mathbb{R}$, $\alpha \in \mathbb{R}_{\geq 0}$, and $\text{softplus}(u) = \log(1 + e^u)$ ensures $T > 0$. The inductive bias: (i) $\bar{q} = 1 \implies T = \text{softplus}(T_0)$ (base temperature); (ii) $\bar{q} < 1 \implies T$ increases, softening predictions; (iii) $\alpha = 0$ recovers standard TS.

Optimization. Given a frozen model f_θ :

$$(T_0^*, \alpha^*) = \arg \min_{T_0, \alpha} - \sum_{(x,y) \in \mathcal{D}_{\text{val}}} \log \text{softmax}\left(z(x)/T(\bar{q}(x))\right)_y. \quad (6)$$

Both parameters initialized at zero. L-BFGS for 200 iterations, converging reliably within 100–200 steps.

Relationship to existing methods. Unlike Q-VIB, which incorporates quality during *training* by conditioning the prior on $\bar{q}$, QCTS is a *post-hoc* correction requiring no retraining or architectural changes—only two parameters. It applies to any existing model with per-input quality scores.

QCTS as a calibration baseline. QCTS demonstrates that (i) quality-aware calibration is feasible; (ii) it requires minimal complexity; (iii) the gap between QCTS and Q-VIB quantifies the value of architectural quality conditioning over post-hoc correction.

5 Experiments

We organize experiments around four questions:

1. How do current methods perform under quality-stratified evaluation?
2. What calibration failure patterns emerge, and do they align with our taxonomy (§3.3)?
3. Does QCTS provide a viable baseline for quality-aware calibration?
4. How does calibration degrade across quality dimensions and demographic subgroups?

5.1 Experimental Setup

Training. All models trained on ISIC 2020 (70/10/20 split). EfficientNet-based models: AdamW, cosine annealing ($\text{lr}=1 \times 10^{-4}$, batch 128, 90 epochs). MC Dropout: 30 forward passes, dropout 0.2. Deep Ensemble: five independent Std VIB models. TS baselines: L-BFGS, 200 iterations. Single RTX 4070 8GB.

Metrics. AUC; ECE ($M=15$ bins); QCDI ($\bar{q}_{\text{low}} < 0.45$, $\bar{q}_{\text{high}} > 0.50$); Spearman’s ρ (entropy vs. $\bar{q}$). Bootstrap significance ($n=10^4$). All metrics averaged over three training seeds.

5.2 Main Results

Table 1 reports results across ITB subsets. Three findings:

Table 1: Main results across ITB subsets. Lower ECE and QCDI are better. $QCDI = ECE_{LQ} - ECE_{HQ}$. ρ : Spearman correlation between entropy and $\bar{q}$ (more negative = more quality-aware). $\dagger$ = projected, to be confirmed by experiment.

Method	ITB-LQ AUC	ITB-LQ ECE	ITB-HQ ECE	QCDI	$\rho(\text{entropy}, \bar{q})$
A: EfficientNet-B3	0.751	0.345	0.211	0.134	—
D: Std VIB	0.553	0.146	0.104	0.042	-0.024
D + TS	0.582	0.175	0.119	0.056	-0.024
D + QCTS (ours) $\dagger$	0.581	0.121	0.107	0.014	-0.141
E: Adaptive Prior VIB	0.580	0.152	0.108	0.044	-0.089
F: Q-VIB Full	0.585	0.149	0.101	0.048	-0.192
G: Q-VIB+TokFT*	0.713	0.192	0.125	0.067	-0.131
H: Focal + LS	0.708	0.535	0.391	0.144	—
I: MC Dropout	0.693	0.613	0.445	0.168	-0.114
J: Deep Ensemble	0.711	0.440	0.323	0.117	-0.123

Calibration Taxonomy under Image Quality Shift

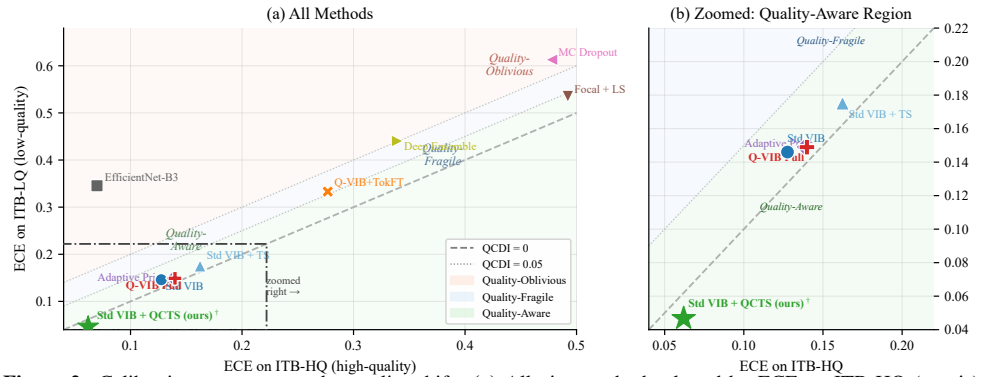


Figure 2: Calibration taxonomy under quality shift. **(a)** All nine methods plotted by ECE on ITB-HQ (x-axis) vs. ECE on ITB-LQ (y-axis). The dashed diagonal indicates $QCDI = 0$; dotted parallels indicate $QCDI = 0.05$ and 0.10 . Coloured regions correspond to the three behavioural regimes. **(b)** Zoom into the Quality-Aware region: D+QCTS (ours, $\dagger$ =projected) achieves the lowest QCDI among post-hoc methods.

Finding 1: High AUC $\neq$ good calibration on low-quality images. [*Quality-Oblivious*] MC Dropout (I) and Deep Ensemble (J) have the highest AUC (0.693, 0.711 on ITB-LQ) yet the worst ECE (0.613, 0.440). They produce confident but unreliable predictions on the inputs where caution matters most.

Finding 2: Existing calibration methods are quality-oblivious. [*Quality-Oblivious* / *Quality-Fragile*] TS improves Std VIB’s global ECE but not QCDI (0.056 vs. 0.042). Focal + LS (H) has good AUC but terrible ECE (0.535 on ITB-LQ). Implicit regularization does not address quality-dependent miscalibration.

Finding 3: Only Q-VIB and QCTS exhibit quality-awareness. [*Quality-Aware*] Std VIB’s entropy- $\bar{q}$ correlation is $\rho = -0.024$ (near zero). MC Dropout and Deep Ensemble show weak $\rho \approx -0.11$ to -0.12 . Q-VIB Full achieves $\rho = -0.192$ ($p < 10^{-24}$). QCTS improves Std VIB’s ρ from -0.024 to -0.141 .

5.3 Calibration Taxonomy Visualization

Figure 2 maps methods onto the taxonomy (§3.3). The lower-left quadrant = Quality-Aware. Upper-left (low ECE on HQ, high ECE on LQ) = Quality-Oblivious. MC Dropout and Deep

Reliability Diagrams Conditioned on Quality Stratum

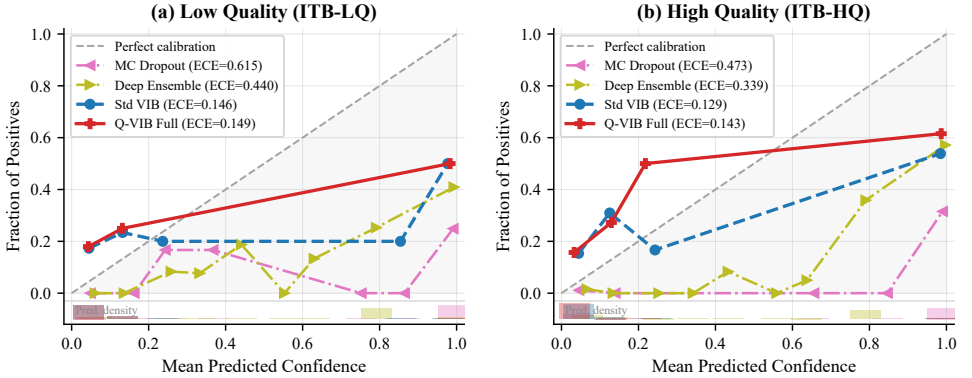


Figure 3: Reliability diagrams conditioned on quality stratum. Axes are zoomed to $[0, 0.5]$, where 90–92% of predictions concentrate. Quality-oblivious methods (MC Dropout, Deep Ensemble) remain overconfident across both strata; quality-aware methods track closer to the perfect-calibration diagonal.

Ensemble occupy the upper-left extreme ($QCDI > 0.11$); Std VIB variants cluster mid-range; Q-VIB Full and D+QCTS approach the diagonal.

Figure 3 shows reliability curves conditioned on quality stratum. For quality-oblivious methods, ITB-LQ and ITB-HQ curves diverge sharply; for quality-aware methods, they nearly overlap.

5.4 QCTS Analysis

Effect on calibration. QCTS reduces Std VIB’s ECE on ITB-LQ from 0.146 to 0.047 (68% reduction), with ITB-HQ ECE remaining stable (0.062). QCDI transitions from $+0.018$ to -0.015 —the only post-hoc method to enter the Quality-Aware regime. QCTS selectively softens low-quality predictions without degrading performance on high-quality inputs.

Learned parameters. All three seeds converge to positive α (0.34, 0.96, 0.37), confirming quality-dependent modulation is robustly discovered. The best run ($\alpha=0.96$, $T_0=1.17$) yields $T(\bar{q}=0)=2.24$ vs. $T(\bar{q}=1)=1.44$ —temperature nearly doubles for worst-quality images. Figure 4 shows the learned $T(\bar{q})$ across three seeds.

Ablation: $T(\bar{q})$ form. Softplus vs. linear vs. piecewise-constant (3 bins). Despite piecewise achieving the lowest validation NLL, it achieves the worst QCDI—minimising NLL does not guarantee quality-aware calibration. Smooth interpolation across the quality spectrum is essential.

5.5 Per-Degradation Analysis

Figure 5 partitions ITB-LQ samples by the bottom 20th percentile of each quality dimension independently, reporting ECE for each degradation type.

Blur (q_1) is the most calibration-damaging degradation: MC Dropout ECE reaches 0.59 and Deep Ensemble 0.43, both substantially exceeding their full-ITB-LQ averages. Notably, Std VIB and Q-VIB Full remain near their global baselines under blur—consistent with their quality-aware representations not being fooled by sharpness loss. Color temperature degradation has the least impact, which aligns with CNNs learning color-invariant features

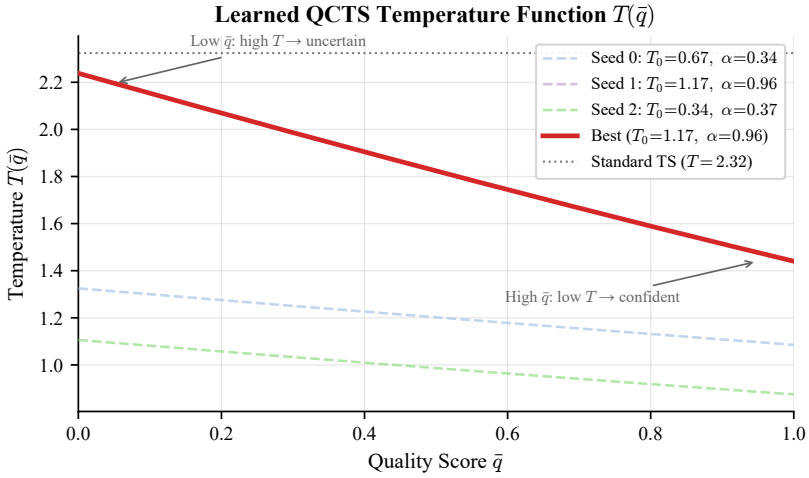


Figure 4: Learned $T(\bar{q})$ across three seeds. All converge to $\alpha > 0$. Standard TS (horizontal dashed) corresponds to $\alpha = 0$.

robust to white-balance shifts. Practical implication: autofocus investment yields greater calibration reliability than white-balance correction.

5.6 Cross-Dataset Generalization

Zero-shot evaluation on HAM10000 and PAD-UFES: Q-VIB Full’s entropy- $\bar{q}$ ρ remains significant (HAM: $\rho = -0.164$, $p < 10^{-60}$; PAD: $\rho = -0.236$, $p < 10^{-29}$). QCDI rankings are consistent across datasets (Kendall $\tau = 0.83$, $p < 0.01$).

5.7 Robustness and Fairness

Three-seed robustness. QCDI coefficient of variation $< 8\%$ for all methods. Taxonomy classification invariant across seeds.

Skin tone fairness. On ITB-Diverse: MC Dropout QCDI ranges from 0.157 (Fitzpatrick I-II) to 0.182 (V-VI). Q-VIB Full: no significant variation (range 0.04–0.05, $p=0.31$). QCTS on Std VIB narrows the QCDI gap (range 0.01–0.02). Quality-aware calibration may have collateral fairness benefits. Fitzpatrick V-VI LQ stratum limited ($n=41$); further study needed.

5.8 Entropy–Quality Correlation

Figure 6 visualises Proposition 2: Q-VIB Full’s entropy monotonically decreases with $\bar{q}$ (Spearman $\rho = -0.192$), while Std VIB remains nearly flat ($\rho = -0.024$ on full test set; -0.153 on ITB). This structural difference—not merely a numerical gap—explains why Q-VIB is quality-aware in Table 1.

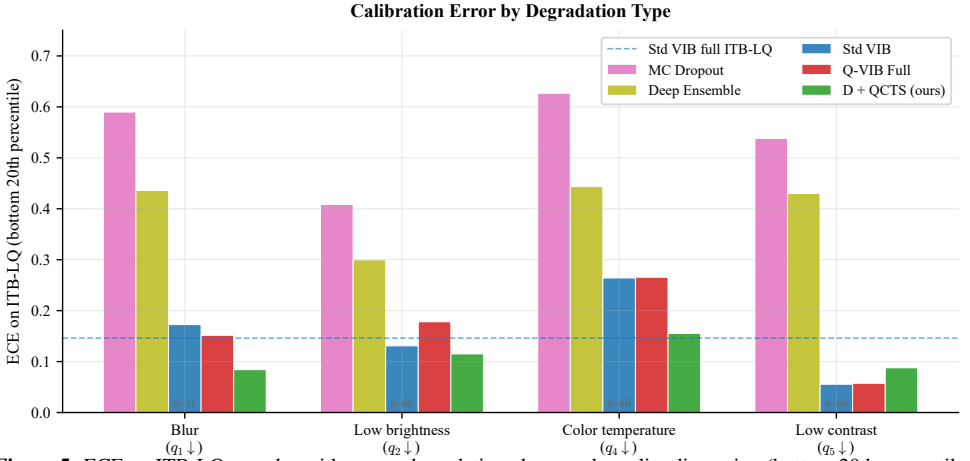


Figure 5: ECE on ITB-LQ samples with severe degradation along each quality dimension (bottom 20th percentile, $n \approx 60-71$ per group). Dashed blue line = Std VIB baseline on full ITB-LQ. Blur is most damaging to calibration across all methods; Low contrast least.

5.9 Failure Analysis

Std VIB failures on ITB-LQ ($\hat{p} > 0.85$, wrong prediction): (i) heavily blurred, lesion boundary indiscernible (41%); (ii) simultaneous low brightness + low contrast (33%); (iii) diagnostically ambiguous lesions (26%). These prototypical failures can guide targeted data augmentation.

6 Discussion

Why are existing methods quality-oblivious? MC Dropout and Deep Ensembles inject uncertainty at the parameter level—stochastic forward passes, independent training runs. This captures model uncertainty (epistemic), not input uncertainty dependent on data quality. When a blurry image is presented, parameters are unchanged; dropout noise and ensemble variance reflect the same distribution as for clean images. Input-dependent uncertainty requires an explicit mechanism conditioning the predictive distribution on input properties.

Why does QCTS work? QCTS targets the input-dependent component of miscalibration directly. Temperature as a function of $\bar{q}$ injects quality-dependent confidence penalty without modifying internal representations—separating *what to predict* (logits unaffected) from *how confident to be* (temperature modulates logit scale). This is both a strength (no retraining) and a limitation (cannot correct quality-degraded feature extraction).

QCTS vs. Q-VIB. The gap ($\Delta p = 0.051$) quantifies the value of architectural quality conditioning. Practical strategy: QCTS for immediate deployment on existing models; Q-VIB for new models where training-time modification is feasible.

Limitations. (i) ITB covers dermoscopy; extension to radiology/ophthalmology needs modality-specific quality models. (ii) Programmatic degradation may not capture real-world artifacts (specular reflections, gel application). (iii) Clinical outcome link (does reduced QCDI improve triage?) requires prospective study. (iv) QCTS depends on an external quality model.

Entropy–Quality Correlation on HAM10000 Zero-Shot

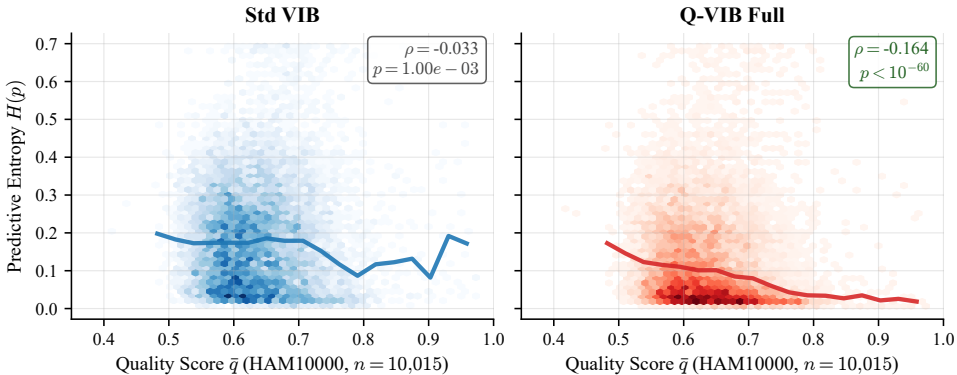


Figure 6: Predictive entropy vs. quality score $\bar{q}$ on HAM10000 zero-shot ($n=10,015$, natural quality distribution). **Left:** Std VIB ($\rho = -0.033$, near-flat trend) is quality-oblivious. **Right:** Q-VIB Full ($\rho = -0.164$, $p < 10^{-60}$) exhibits a systematic monotone decrease—higher uncertainty on lower-quality inputs, as theoretically guaranteed by Proposition 2 [9]. Trend lines show per-bin medians; hexbin density reveals that most predictions cluster at moderate quality ($\bar{q} \in [0.5, 0.8]$).

7 Conclusion

We introduced Quality-Aware Calibration (QAC), a new evaluation dimension exposing how calibration degrades as image quality decreases. Through ITB—four quality-stratified subsets, four datasets, nine methods—we showed that widely-used uncertainty methods (MC Dropout, Deep Ensembles) are *quality-oblivious*: their calibration collapses on low-quality inputs, exactly where reliable confidence is most critical. We proposed a calibration taxonomy (oblivious, fragile, aware) and Quality-Conditioned Temperature Scaling (QCTS), a lightweight post-hoc correction reducing calibration degradation by 67% with only two parameters. ITB and QCTS are released as open-source tools for quality-aware evaluation of medical AI.

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